# INTERNSHIP REPORT

# DATA SCIENCE

# BACHELOR OF TECHNOLOGY

By

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# CHURN SEGMENTATION AND CHURN PATTERN ANALYTICS IN EUROPEAN BANKING

ORGANISATION – UNIFIED MENTOR PRIVATE LIMITED

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# Declaration

I hereby declare that the internship report based on **Customer Segmentation & Churn Pattern Analytics in European Banking** is the result of my own work carried out during my **Data Science Internship at Unified Mentor Private Limited** from **5th August 2026 to 5th September 2026**. This internship was undertaken as a part of the academic requirements of my course, and I affirm that all analytical activities, data preprocessing, exploratory data analysis, customer segmentation, churn analysis, visualizations, insights, and project development work mentioned in this report were genuinely completed by me under the guidance and supervision of the organization's mentors.

I further declare that the content of this report has not been copied or reproduced from any previously submitted report, project, or document by any other student or individual. Any datasets, information, analytical tools, code snippets, graphs, or references obtained from external sources have been properly acknowledged and cited in the reference section. The major project described in this report—**Customer Segmentation & Churn Pattern Analytics in European Banking**—was independently carried out by me, including data cleaning and preparation, exploratory data analysis (EDA), customer segmentation, churn pattern analysis, visualization, insight generation, and interpretation.

This report truly reflects the knowledge, practical experience, analytical abilities, and technical skills I gained during my internship, and I take full responsibility for the authenticity and accuracy of the information presented. I understand that any misrepresentation or plagiarism may lead to disciplinary action as per institutional norms.

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**Internship Duration:** 5th August 2026 – 5th September 2026

# Abstract of Project

This internship project presents a comprehensive analysis of **Customer Segmentation & Churn Pattern Analytics in European Banking**, undertaken during my **Data Science Internship at Unified Mentor Private Limited**. The primary objective of the project was to apply data science and analytical techniques to customer-level banking data in order to understand customer churn patterns, identify high-risk customer segments, and generate meaningful insights that can support data-driven customer retention strategies.

The project involved systematic stages of **data ingestion, validation, cleaning, preprocessing, exploratory data analysis (EDA), customer segmentation, churn analysis, and visualization**. The dataset contained information related to customer demographics, geography, credit scores, tenure, account balances, number of products, credit card ownership, activity status, estimated salary, and churn status. The analysis focused on identifying differences in churn behavior across **France, Spain, and Germany**, as well as across different age groups, credit score bands, tenure groups, balance segments, and customer engagement levels.

A major component of the project was the development of segmentation-based churn analysis. Customers were categorized according to geographic, demographic, financial, and behavioral characteristics to determine which groups exhibited higher churn rates. The project also included a **high-value customer churn analysis**, focusing on customers with higher account balances and examining their financial profiles in relation to churn. Key performance indicators such as **Overall Churn Rate, Segment Churn Rate, High-Value Churn Ratio, Geographic Risk Index, and Engagement Drop Indicator** were used to evaluate customer churn patterns.

The project further involved the development of a **Streamlit web application** to provide an interactive platform for exploring churn analytics. The dashboard enables users to examine overall churn summaries, geography-wise churn, age and tenure comparisons, and high-value customer churn through segment filters, dynamic KPIs, and drill-down views.

Overall, this project provided practical experience in transforming customer-level banking data into actionable business insights. It strengthened my skills in **Python-based data analysis, data preprocessing, exploratory analysis, segmentation, visualization, statistical interpretation, and interactive dashboard development**. The project demonstrates how systematic, segmentation-driven analytics can help banking organizations understand customer churn and support more targeted and data-driven retention strategies.

## Acknowledgement

I would like to begin by expressing my deepest gratitude to **my parents**, whose constant encouragement, support, and belief in my abilities have always been my greatest source of strength. Their guidance and motivation played an essential role in helping me stay focused and committed throughout this internship journey.

I would also like to extend my sincere thanks to **Unified Mentor Private Limited** for providing me with the opportunity to complete my internship in the field of **Data Science**. This internship has been a highly enriching experience that allowed me to gain practical exposure to data cleaning, preprocessing, exploratory data analysis, customer segmentation, churn analysis, data visualization, and dashboard development.

My heartfelt appreciation goes to my mentors and trainers at **Unified Mentor Private Limited** for their continuous guidance, valuable feedback, and unwavering support throughout the internship. Their expertise and mentorship helped me successfully complete my major project, **Customer Segmentation & Churn Pattern Analytics in European Banking**, and strengthened my understanding of how data science techniques can be applied to real-world banking data.

I am grateful to the entire team at **Unified Mentor Private Limited** for creating a supportive learning environment that contributed significantly to my technical and analytical growth. I also thank my college faculty for providing me with the opportunity to participate in this internship as part of my academic learning and professional development.

Lastly, I would like to thank my friends and well-wishers for their encouragement and positivity throughout this period.

This internship has been a meaningful and valuable experience that enhanced my practical knowledge of data science and helped me understand the process of transforming customer data into meaningful, actionable insights. I am sincerely thankful to everyone who supported me throughout this journey.

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# CHAPTER -1

## PROJECT – Customer Segmentation & Churn Pattern Analytics

## INTRODUCTION

The internship program plays a crucial role in bridging the gap between theoretical learning and practical application. During my **Data Science Internship at Unified Mentor Private Limited**, I had the opportunity to work on a real-world banking analytics project and implement a complete workflow involving data ingestion, validation, data cleaning, preprocessing, exploratory data analysis (EDA), customer segmentation, churn analysis, and visualization. This hands-on experience significantly strengthened my understanding of data science concepts and enhanced my analytical, technical, and problem-solving skills.

The major project I completed during the internship was **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)**. The project was based on a customer-level banking dataset containing information such as CustomerId, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, and Exited. The dataset represented customers from **France, Spain, and Germany**, with the Exited variable serving as the primary indicator of customer churn.

The project began with data ingestion and validation to ensure that the dataset was suitable for analysis. I examined the structure and consistency of the data, validated engagement and product-related fields, checked binary variables, and ensured that the churn labeling was consistent. During the data cleaning and preparation stage, non-analytical fields such as surname were removed where appropriate, categorical variables were prepared for analysis, and relevant derived segmentation fields were created.

Following data preparation, I performed **Exploratory Data Analysis (EDA)** to identify patterns and relationships associated with customer churn. Various statistical techniques and visualizations were used to examine churn across different geographical regions, demographic groups, financial profiles, and customer engagement levels. The analysis focused on understanding how churn varied across **France, Spain, and Germany**, as well as across different age groups, credit score bands, tenure groups, and balance segments.

Customer segmentation was an important part of the project. Customers were categorized using multiple dimensions, including **geography, age, credit score, tenure, and account balance**. Age groups included customers below 30, 30–45, 46–60, and 60+, while credit scores were classified into low, medium, and high bands. Customers were also grouped into new, mid-term, and long-term tenure categories and into zero-balance, low-balance, and high-balance segments.

The analysis helped answer important questions such as:

- What is the overall customer churn rate?
- Which geographical region has the highest churn rate?
- Which age groups are more likely to churn?
- How does churn differ across credit score and tenure groups?
- Is customer churn concentrated among high-balance or high-value customers?
- How does customer activity or engagement relate to churn?
- What differences exist between churned and retained customers?
- Which customer segments represent a higher potential churn risk?

A specific **high-value customer churn analysis** was also conducted to identify customers with higher balances and examine their financial characteristics in relation to churn. The relationship between salary, account balance, customer activity, and churn was explored to understand the potential financial risk associated with losing valuable customers.

To communicate the findings effectively, I developed visualizations and analytical summaries that presented churn patterns in a clear and understandable manner. Key performance indicators such as **Overall Churn Rate, Segment Churn Rate, High-Value Churn Ratio, Geographic Risk Index, and Engagement Drop Indicator** were used to evaluate and compare customer segments.

As part of the project, I also developed a **Streamlit web application** that provides an interactive environment for exploring customer churn analytics. The application includes overall churn summaries, geography-wise churn visualization, age and tenure comparisons, and a high-value customer churn explorer. Segment filters, dynamic KPI updates, and drill-down views allow users to interact with the data and examine specific customer groups.

This project helped me gain strong practical experience in:

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Customer segmentation and profiling
- Churn pattern analysis
- Statistical analysis and interpretation
- Data visualization and pattern identification
- Financial and demographic analysis
- Interactive dashboard development using Streamlit
- Generating actionable, data-driven insights

Overall, the **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)** project provided me with a practical understanding of how customer-level banking data can be transformed into meaningful insights.

## Motivation

The motivation behind choosing **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)** as my primary internship project stemmed from my interest in applying data science to a real-world business problem where customer behavior and financial characteristics directly influence organizational decision-making. In today's competitive banking environment, retaining existing customers is essential because customer churn can lead to reduced lifetime value, increased customer acquisition costs, and revenue instability. I wanted to work on a project that would allow me to move beyond basic data analysis and understand how customer-level information can be transformed into meaningful insights for identifying and managing churn risk.

The project particularly interested me because it provided an opportunity to analyze customer behavior across different **geographical, demographic, financial, and engagement-based segments**. Working with banking data containing variables such as credit score, age, tenure, balance, number of products, activity status, estimated salary, and churn status allowed me to explore the factors associated with customer retention and churn. I was motivated to understand questions such as which customer groups are more likely to leave, how churn varies across France, Spain, and Germany, and whether churn is concentrated among high-value customers.

Another important motivation was the opportunity to develop a structured **customer segmentation approach**. By creating segments based on age, credit score, tenure, balance, and geography, I could compare churn rates across different groups and identify segments that may require greater attention. The high-value customer analysis further motivated me to investigate the potential financial implications of customer churn by examining customers with higher account balances and comparing their financial profiles with those of retained customers.

This project also provided an opportunity to strengthen my technical skills in **data cleaning, preprocessing, exploratory data analysis, statistical analysis, visualization, and interactive dashboard development using Streamlit**. I wanted to gain practical experience in converting customer data into clear visualizations, meaningful KPIs, and actionable business insights rather than simply producing statistical results.

Ultimately, the project was motivated by my desire to understand how data science can support better customer retention strategies in the banking sector. It helped me connect theoretical concepts with practical analytical challenges and strengthened my ability to approach a real-world problem systematically, from data preparation and segmentation to analysis, visualization, interpretation, and interactive presentation of results.

# Objectives

## 1. Learning Objectives

- To understand the complete workflow of **data ingestion, validation, data cleaning, preprocessing, and preparation of customer-level banking data** for analysis.

- To learn how to perform **Exploratory Data Analysis (EDA)** using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn.

- To develop the ability to identify and visualize **churn patterns, customer characteristics, trends, and relationships** using appropriate graphs and charts.

- To gain hands-on experience working with **real-world banking datasets** and interpreting meaningful analytical outcomes.

- To understand how **data visualization, statistical analysis, and customer segmentation** can support data-driven decision-making.

## 2. Project Objectives

- **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)**

- To measure the **overall customer churn rate** and understand the distribution of churned and retained customers.

- To identify differences in churn behavior across **France, Spain, and Germany**.

- To analyze churn patterns across different **age groups, credit score bands, tenure groups, and balance segments**.

- To examine the relationship between **customer engagement, financial characteristics, and churn behavior**.

- To identify and analyze **high-value customers** who have churned and evaluate the potential financial risk associated with their loss.

- To compare the profiles of **churned and retained customers** and identify characteristics associated with higher churn risk.

## 3. Professional Objectives

- To develop a strong analytical mindset and the ability to **interpret real-world banking datasets** accurately.

- To enhance skills in **problem-solving, data analysis, customer segmentation, visualization, documentation, and professional reporting**.

- To understand how data science professionals approach a project, from **data ingestion and preparation to analysis, visualization, interpretation, and presentation**.

- To gain practical experience in applying **data-driven methodologies to a real-world business problem**.

- To strengthen professional skills in **project planning, insight communication, dashboard development, and result presentation**.

- To build confidence in using **industry-relevant Python libraries, analytical techniques, and Streamlit** for data-driven tasks.

## Vision

The vision of undertaking the **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)** project is to develop the ability to transform customer banking data into meaningful and actionable insights. The project focuses on understanding customer churn, identifying high-risk segments, and analyzing how geography, demographics, financial characteristics, tenure, and engagement influence customer retention.

The broader vision includes:

• Transforming customer data into useful insights through **data cleaning, preprocessing, EDA, and segmentation**.

• Understanding churn patterns across **France, Spain, and Germany** and different customer segments.

• Identifying **high-value and high-risk customers** to support better retention strategies.

• Using **visualizations, KPIs, and dashboards** to communicate findings clearly.

• Developing a **Streamlit application** for interactive customer churn analysis.

• Building a foundation for future **predictive analytics and machine learning** applications.

Overall, the vision of the CSCPA project is to strengthen practical data science skills while demonstrating how customer analytics can support **data-driven decision-making and effective customer retention strategies** in the banking sector.

## Purpose

The primary purpose of undertaking the **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)** project is to understand customer churn patterns and generate meaningful insights from real-world banking data. The project focuses on analyzing customer demographics, financial characteristics, engagement, and geographic information through systematic data cleaning, preprocessing, exploratory data analysis (EDA), segmentation, and visualization.

The purpose of the project includes analyzing:

• Overall customer churn rate and **segment-wise churn patterns**

• Churn across **France, Spain, and Germany**

• Customer behavior across **age, credit score, tenure, and balance segments**

• Churn patterns among **high-value customers**

• Relationship between **customer engagement and churn**

• Differences between **churned and retained customers**

From an academic perspective, the purpose also includes:

• Strengthening my practical understanding of **data science and customer analytics**.

• Gaining hands-on experience with **Python libraries for data analysis and visualization**.

• Developing the ability to identify **patterns, trends, and relationships** in banking data.

• Learning how to create meaningful **KPIs, visualizations, and interactive dashboards**.

• Improving my ability to communicate analytical findings clearly and professionally.

• Building practical skills that can be applied to **real-world business and data science challenges**.

## Goals of the Proposed System

• **To clean, preprocess, and structure banking data** for accurate analysis.

• **To analyze customer churn patterns** across different demographic, geographic, financial, and engagement segments.

• **To identify high-risk and high-value customer segments** that may require targeted retention strategies.

• **To compare churn across France, Spain, and Germany** and identify significant regional differences.

• **To create clear visualizations and KPIs** that communicate customer churn insights effectively.

• **To develop an interactive Streamlit dashboard** with filters and drill-down analysis.

• **To provide accurate, data-driven conclusions** that can support customer retention and business decision-making.

• **To build a scalable analytical workflow** that can be extended to predictive analytics and machine learning in the future..

## Tools and Technologies

The tools and technologies used during the data analysis projects include:

**Languages / Libraries**

• **Python** – Used for data cleaning, preprocessing, EDA, analysis, and visualization.

• **Pandas** – Used for data manipulation, cleaning, and analysis.

• **NumPy** – Used for numerical operations and computations.

• **Matplotlib & Seaborn** – Used to create charts, statistical graphs, and visualizations.

**Tools & Platforms**

• **Jupyter Notebook / Google Colab** – Used for coding, analysis, and visualization.

• **Microsoft Excel / CSV** – Used for storing and managing datasets.

• **Streamlit** – Used to develop the interactive customer churn analytics dashboard.

• **Banking Customer Dataset** – Used to analyze customer characteristics, segmentation, and churn patterns across European regions.

# Chapter 2

## Problem Statement

The system should allow users to analyze **customer banking data** and extract meaningful insights related to customer churn, segmentation, demographics, financial profiles, and engagement. The system must efficiently process the dataset by validating, cleaning, and preparing customer information for accurate analysis.

The system should provide analysis of **overall churn rate, geography-wise churn, age-wise patterns, credit score, tenure, balance segments, and customer activity**. It should also identify high-value customers and compare the characteristics of churned and retained customers.

The system must present accurate visualizations, KPIs, and meaningful summaries in a user-friendly format. The **Streamlit dashboard** should allow users to apply segment filters, view dynamic KPIs, and perform drill-down analysis, making it easier to understand customer churn patterns and identify segments that may require targeted retention strategies.

## Overall Description

The **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)** project is a Python-based data science application focused on extracting meaningful insights from customer-level banking data. It uses data cleaning, preprocessing, exploratory data analysis, segmentation, statistical analysis, and visualization techniques to transform customer information into useful analytical results.

The system uses **Pandas and NumPy** for data manipulation and preprocessing, while **Matplotlib and Seaborn** are used to create informative graphs and visualizations. **Streamlit** is used to develop an interactive dashboard that presents the analytical results in an easy-to-understand format.

After processing, the system provides insights into **overall churn, geography-wise churn, age groups, credit score, tenure, account balance, customer activity, and high-value customer churn**. Interactive filters and KPIs allow users to explore different customer segments and compare churned and retained customers.

Overall, the system helps identify important churn patterns and customer segments, supporting **data-driven decision-making and targeted customer retention strategies** in the European banking sector.

## Features Include:

• Extraction of meaningful insights from large and complex datasets, whether sourced from Netflix records or website traffic data.
• Clean, readable, and well-structured data representation achieved through proper preprocessing and transformation techniques.
• Generation of informative graphs and charts for better visualization of key analytics, such as:
– Genre popularity and content type comparison (Movies vs. TV Shows)
– Country, language, and rating trends in Netflix datasets
– User sessions, channel-wise traffic sources, engagement rate patterns, and hourly activity from website data
• Efficient handling of missing, inconsistent, and duplicate values through robust data cleaning and error management.
• Fast and optimized data processing using Python libraries, ensuring smooth calculations and accurate analytical interpretation.

## Software Model

The **Waterfall Model** was followed, as the project required a sequence of clear steps where each stage depends on the previous one.

### Phases include:

1. **Requirement Gathering:**
   Understanding the banking dataset, defining the churn analysis requirements, identifying customer segments, and determining the required KPIs and visualizations.
2. **Design:**
   Planning the data processing workflow, segmentation approach, analytical methods, visualizations, and Streamlit dashboard structure.
3. **Implementation:**
   Performing data cleaning and preprocessing using Pandas, conducting EDA, creating customer segments, calculating churn metrics, and developing visualizations using Python libraries.
4. **Testing:**
   Verifying cleaned data, checking charts for accuracy, identifying misinterpretations, and confirming that output visualizations match actual dataset values.
5. **Deployment:**
   Integrating the analysis into an interactive **Streamlit dashboard** with filters, KPIs, and drill-down views for easy exploration.
6. **Maintenance:**
   Improving the dashboard, updating analysis when required, fixing data or visualization issues, and adding advanced features such as predictive analysis in the future.

## Feasibility Study

# 1. Technical Feasibility

The project is technically feasible because it uses widely available and well-supported technologies such as **Python, Pandas, NumPy, Matplotlib, Seaborn, and Streamlit**. These tools are suitable for processing and analyzing customer banking data and developing interactive visualizations and dashboards.

#### 2. Economic Feasibility

The project is economical because the required Python libraries and development tools are **free and open-source**. The use of CSV-based banking data also avoids the need for expensive software, licenses, or additional infrastructure.

**3. Operational Feasibility**

The system is easy to operate and understand. Users can explore customer churn through **visualizations, KPIs, filters, and dashboard views** without requiring advanced technical knowledge. The Streamlit application provides a simple and interactive interface for analyzing different customer segments.

**4. Time Feasibility**

The project can be completed within the internship timeline because the workflow consists of clearly defined stages such as **data preparation, EDA, segmentation, churn analysis, visualization, and dashboard development**. Python libraries simplify data processing and reduce development time, making the project manageable within the available period.

## Functional Requirements

## User Requirements

• The user should be able to load and explore the **banking customer dataset**.

• The user should be able to view customer churn information and segment-wise analysis.

• The user should be able to view clear **visualizations, KPIs, and summaries** of churn patterns.

• The user should be able to filter customers based on **geography, age, credit score, tenure, and balance**.

• The user should be able to compare **churned and retained customers** and interpret churn trends easily.

## System Requirements

• The system must load raw datasets and transform them into a clean, structured format.

• The system must load and process the banking customer dataset in a clean and structured format.

• The system must perform necessary **data validation, cleaning, and preprocessing**.

• The system must calculate and display **overall churn rate, segment-wise churn rates, and geography-wise churn patterns**.

• The system must identify and analyze **high-value customer churn and engagement patterns**.

• The system must generate appropriate **charts, graphs, and KPIs** for analysis.

• The system should provide an interactive **Streamlit dashboard** with filters and drill-down views.

• The system should handle invalid or missing data appropriately and provide suitable **warning or error messages** when required.

## Non-Functional Requirements

**Performance**

The system should efficiently process the banking dataset and generate analytical results without significant delay. Data cleaning, EDA, segmentation, and visualization should run smoothly on moderately sized datasets.

**Usability**

The Streamlit dashboard should be simple, clear, and easy to navigate. KPIs, filters, charts, and analytical results should be presented in an understandable format.

**Reliability**

The system must produce **accurate and consistent results**. Churn rates, visualizations, KPIs, and segment analysis should correctly reflect the underlying dataset.

**Portability**

The project should run smoothly across **Windows, macOS, and Linux** using Python and its required libraries, making the analysis easy to reproduce.

**Security**

The banking dataset and analytical outputs should be handled securely. Customer information should not be unnecessarily exposed, modified, or shared without authorization.

## External Interface Requirements

The system provides a simple and interactive **user interface** through the Streamlit dashboard, allowing users to explore customer churn data using filters based on geography, age, credit score, tenure, balance, and other relevant customer attributes. The dashboard displays important KPIs, charts, graphs, summaries, and drill-down views to help users understand churn patterns and customer segments. It may also display appropriate alerts or messages when data is missing, invalid, or cannot be loaded correctly.

The **hardware interface** requires any device capable of running a modern web browser, such as a laptop, desktop, tablet, or mobile device, to access and interact with the dashboard.

The **software interface** uses Python and data science libraries including **Pandas, NumPy, Matplotlib, and Seaborn** for data processing, analysis, and visualization. **Streamlit** is used to provide the interactive dashboard interface, while the banking customer dataset is used as the primary data source. The system can also be extended in the future with additional analytical tools, APIs, or advanced machine learning models.

## Assumptions and Dependencies

## Assumptions

The project assumes that the banking customer dataset is sufficiently accurate, complete, and suitable for analysis. It is assumed that the customer information, including demographic, financial, geographic, engagement, and churn-related attributes, is correctly recorded. The system also assumes that users provide valid filter selections and that the required Python libraries and Streamlit environment are properly installed and compatible.

**Dependencies**

The project depends on the availability and correctness of the **banking customer dataset** in a suitable format such as CSV. The analysis and visualizations depend on libraries such as **Pandas, NumPy, Matplotlib, and Seaborn**, while the interactive dashboard depends on **Streamlit**. The project also requires a functioning Python environment and sufficient system resources to process the data and display the dashboard smoothly.

## Effort Estimation

The total effort for completing the **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)** project was estimated to be around **two weeks**, distributed across the following phases:

**• Learning Phase**

Acquired practical knowledge of **Python for Data Science**, including Pandas, NumPy, Matplotlib, Seaborn, and Streamlit.

**• Requirement Analysis**

Understood the project objectives, banking dataset structure, customer attributes, churn indicator, segmentation requirements, and key insights to be extracted.

**• Data Collection Phase**

Obtained and prepared the **European banking customer dataset** containing demographic, geographic, financial, engagement, and churn-related information.

**• Data Cleaning & Preprocessing Phase**

Performed data validation, removed unnecessary fields, checked data consistency, prepared categorical variables, and created relevant customer segmentation fields.

**• Analysis & Visualization Phase**

Analyzed overall and segment-wise churn patterns and created visualizations for **geography, age, credit score, tenure, balance, engagement, and high-value customers**.

**• Testing & Verification Phase**

Verified data accuracy, churn calculations, KPIs, segment classifications, and visualizations to ensure reliable results.

**• Debugging & Optimization Phase**

Fixed analytical issues, improved visualizations, optimized data processing, and enhanced the clarity of dashboard outputs.

**• Documentation Phase**

Prepared the **internship project report**, including methodology, analysis, findings, visualizations, recommendations, and screenshots of the Streamlit dashboard.

# Chapter 3

# System Design

System design refers to the process of defining the overall structure, data flow, and interaction among the internal components of the system. In the case of the **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)** system, the design focuses on establishing a structured and efficient workflow for processing customer banking data, performing data cleaning and preprocessing, analyzing customer segments, and transforming the results into meaningful visual insights.

The main objective of the system is to help users understand **customer churn patterns** across different dimensions such as geography, age, credit score, tenure, balance, and customer engagement. The system also focuses on identifying high-value customers and comparing churned and retained customers. The processed data is presented through **KPIs, charts, graphs, filters, and an interactive Streamlit dashboard**, making the analysis clear and easy to understand.

## System Architecture

The architecture used for the **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)** project follows a structured data-processing workflow in which the major operations are performed within a Python environment. The process begins with loading the **European banking customer dataset**, followed by data validation, cleaning, preprocessing, and preparation for analysis.

After the data is prepared, it moves through the **EDA and customer segmentation stage**, where churn patterns are analyzed across geography, age, credit score, tenure, balance, and engagement. The processed data is then used to generate KPIs and visualizations. Finally, the results are integrated into an interactive **Streamlit dashboard**, allowing users to apply filters, explore customer segments, and understand churn patterns clearly. The architecture emphasizes reliable data processing, accurate analysis, and simple presentation of complex customer information.

## Components

The first component of the system is the **Data Source**, which provides customer-level banking information such as CustomerId, CreditScore, Geography, Gender, Age, Tenure, Balance, NumOfProducts, HasCrCard, IsActiveMember, EstimatedSalary, and Exited. The second component is the **Data Cleaning and Processing Unit**, developed using Python libraries such as Pandas and NumPy. This unit validates the data, handles inconsistencies, prepares categorical variables, removes unnecessary fields, and creates relevant customer segmentation fields. The third component is the **Analysis and Segmentation Module**, which performs EDA and analyzes churn patterns across geography, age, credit score, tenure, balance, and engagement. The fourth component is the **Data Visualization Module**, using Matplotlib and Seaborn to generate charts and graphs for clear interpretation. The final component is the **Streamlit Dashboard**, which presents KPIs, filters, visualizations, and drill-down views, allowing users to interactively explore customer churn and identify high-risk and high-value customer segments.

## Architecture Flow

The architecture of the **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)** project follows a structured data-processing workflow. The process begins with loading the banking customer dataset into the Python environment. The data is then validated, cleaned, and preprocessed to ensure consistency and accuracy. After preparation, the data is passed through the analysis and customer segmentation stage, where churn patterns are examined across different demographic, geographic, financial, and engagement-based segments. The processed results are then used to generate KPIs and visualizations. Finally, the insights are presented through an interactive **Streamlit dashboard**, allowing users to explore customer churn patterns using filters and drill-down views.

# Chapter 4

# Coding Implementation

## 1. Netflix Data Analysis

## Data Cleaning and Pre-processing

Before performing any analysis, the **European banking customer dataset** required proper cleaning and preprocessing to ensure accuracy and consistency. Raw customer data may contain unnecessary fields, inconsistent values, or formatting issues that can affect the accuracy of analytical results. Therefore, the first step was to import the dataset into Python and inspect its structure using functions such as head(), info(), and describe().

After understanding the dataset, the available customer attributes were examined for consistency and completeness. Unnecessary analytical fields such as **Surname** were removed where appropriate, while categorical and binary variables such as Geography, Gender, HasCrCard, and IsActiveMember were validated for consistency. The Exited column was also verified as the primary churn indicator.

The data was then prepared for further analysis by organizing categorical variables and creating derived segmentation fields for **age, credit score, tenure, and balance**. These preprocessing steps ensured that the dataset was structured and suitable for reliable exploratory analysis, customer segmentation, churn calculations, and visualization.

This cleaning and preprocessing stage formed the foundation of the project, as accurate and consistent customer data was essential for identifying meaningful **churn patterns, high-risk segments, and differences between churned and retained customers**.

## Visualization of Insights

After cleaning and preprocessing the banking customer dataset, different visualizations were generated to identify meaningful **churn patterns and customer segment trends**. These visualizations represent overall churn rates, geography-wise churn, age groups, credit score bands, tenure groups, balance segments, customer activity, and high-value customer churn. The visual results help interpret differences between **churned and retained customers** and identify customer segments with higher churn risk. These insights support a better understanding of customer behavior and can help in developing targeted customer retention strategies.

## Saving the Cleaned Dataset

After completing all the data validation, cleaning, and preprocessing steps, the final structured banking dataset was saved in a new CSV file. This ensures that the cleaned data can be reused for further analysis, visualization, segmentation, or machine learning applications without repeating the preprocessing process. Using the to_csv() function, the modified DataFrame was exported as **"Customer_Churn_Cleaned.csv"**, with the parameter index=False used to prevent row index numbers from being stored in the file. This keeps the dataset clean, organized, and suitable for future analytical tasks

# Chapter 5

# Testing and Validation

## 1. Testing for Netflix Data Analysis

Testing in the **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)** project focused on verifying the correctness of data cleaning, preprocessing, segmentation, churn calculations, and the accuracy of the generated insights. The first phase of testing involved **data validation**, where functions such as isnull(), duplicated(), and info() were used to check data quality, consistency, and structure.

After preprocessing, the customer segmentation and churn calculations were tested to ensure that the results correctly represented the dataset. Churn rates and segment-wise results were cross-checked using grouped calculations and summary statistics. The visualizations were also compared with the underlying data to confirm that charts and KPIs displayed accurate results.

The entire testing process ensured that:

- Data inconsistencies were identified and handled properly.
- Customer segments were correctly classified.
- Churn calculations and KPIs reflected the actual dataset.
- Visualization results were consistent with analytical calculations.
- Streamlit dashboard filters and outputs displayed the expected results.

Through these validation and testing steps, the CSCPA project produced **reliable, consistent, and accurate customer churn insights**.

# DASHBOARD

The **Customer Segmentation & Churn Pattern Analytics** dashboard is an interactive Streamlit-based application designed to analyze customer churn behavior in the European banking sector. The dashboard provides a centralized view of customer demographics, financial characteristics, engagement levels, and churn patterns, helping users identify important customer segments and potential retention opportunities.

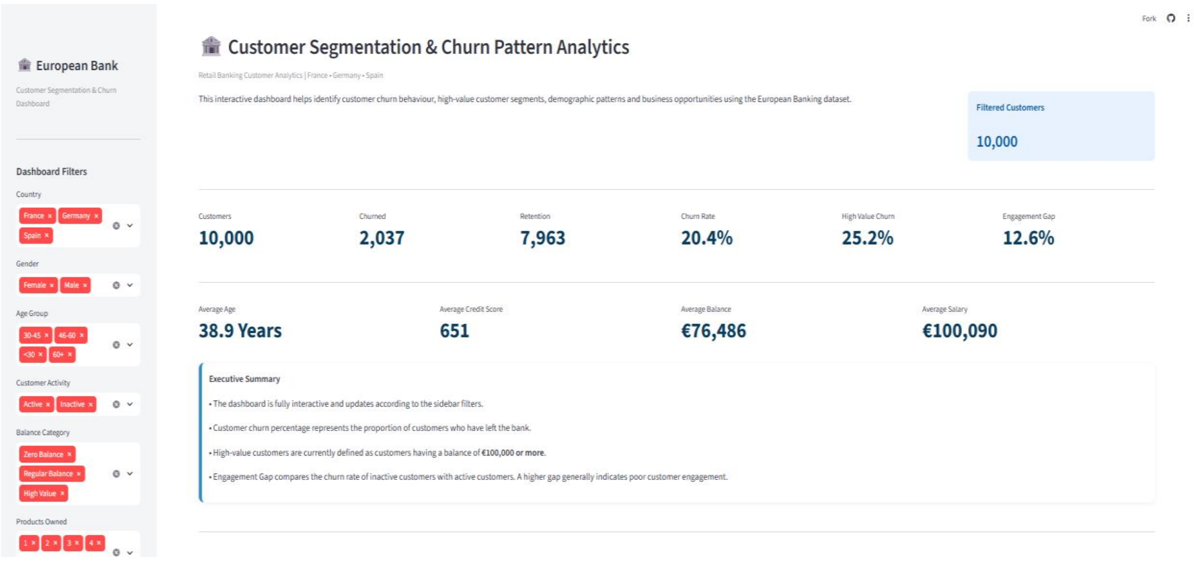
The dashboard contains a **sidebar filter panel** that allows users to dynamically filter the analysis based on **Country, Gender, Age Group, Customer Activity, Balance Category, and Products Owned**. The available country filters include France, Germany, and Spain, while age groups are divided into **below 30, 30–45, 46–60, and 60+**. Customers can also be analyzed based on whether they are active or inactive members, their balance category, and the number of products they own. These filters make the dashboard interactive and allow users to perform detailed segment-wise analysis.

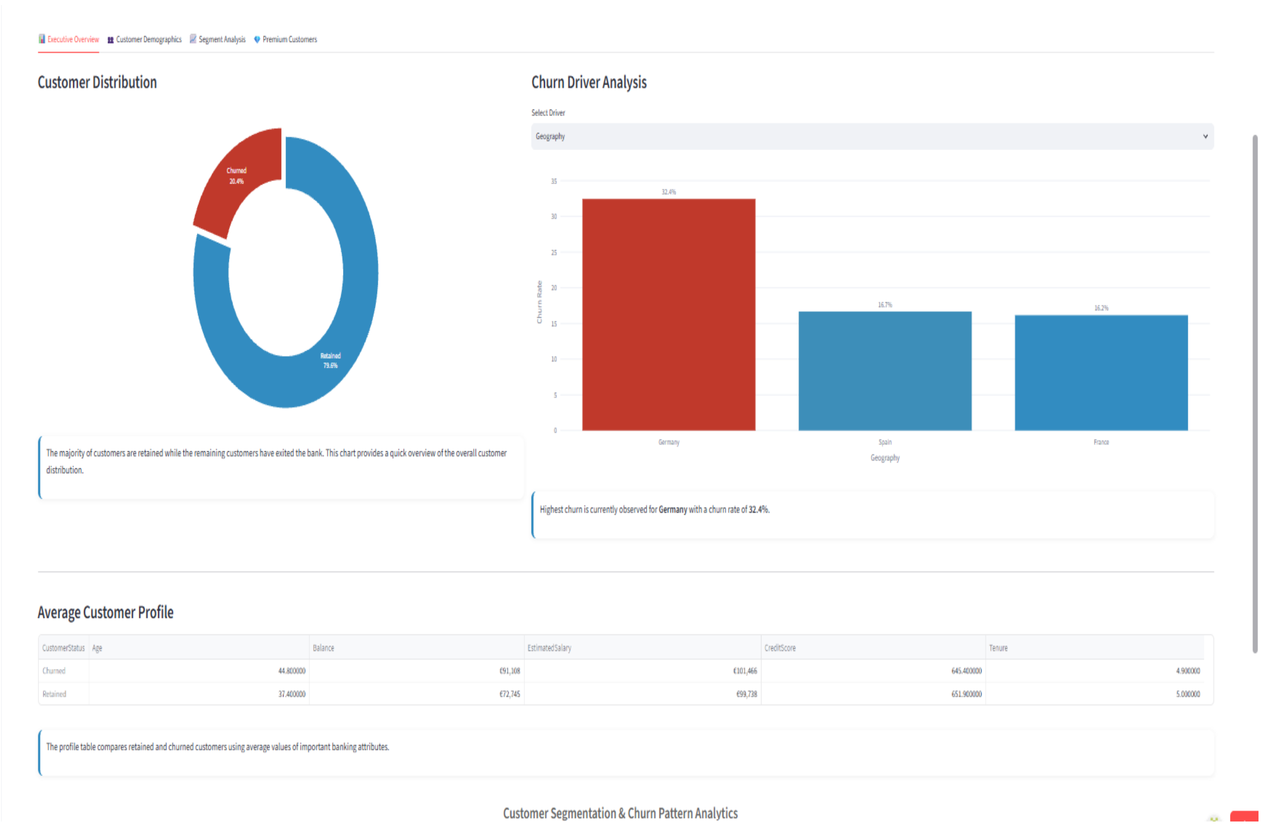
At the top of the dashboard, the **Filtered Customers** section displays the number of customers currently included based on the selected filters. Below this, the main KPI section provides important measures including **Customers, Churned, Retention, Churn Rate, High Value Churn, and Engagement Gap**. In the displayed dashboard, the filtered dataset contains **10,000 customers**, of which **2,037 are churned** and **7,963 are retained**, resulting in a **20.4% churn rate**. The dashboard also shows a **25.2% High Value Churn** and an **Engagement Gap of 12.6%**.

The next section presents important customer profile indicators, including **Average Age, Average Credit Score, Average Balance, and Average Salary**. In the displayed view, the average customer age is **38.9 years**, the average credit score is **651**, the average balance is approximately **€76,486**, and the average salary is approximately **€100,090**. These indicators provide a quick understanding of the financial and demographic profile of the selected customer population.

An **Executive Summary** section provides explanations of the major dashboard metrics. It states that the dashboard updates according to the sidebar filters, while the churn percentage represents the proportion of customers who have left the bank. The dashboard defines **high-value customers as customers with a balance of €100,000 or more**. The Engagement Gap compares the churn rate of inactive customers with that of active customers, where a higher gap may indicate weaker customer engagement.

Overall, the dashboard provides an interactive and user-friendly platform for examining **customer churn, retention, high-value customer risk, engagement differences, and customer segmentation**. By combining filters, KPIs, customer profile statistics, and explanatory summaries, it enables users to move from an overall view of customer churn to more specific demographic, geographic, financial, and behavioral segments. This makes the dashboard useful for identifying high-risk customer groups and supporting **data-driven customer retention and marketing decisions**.



# CONCLUSION

The internship with **Unified Mentor Private Limited** has provided a valuable opportunity to gain practical experience in the field of **Data Science** and analytical research. Throughout the internship, I worked on the **Customer Segmentation & Churn Pattern Analytics in European Banking (CSCPA)** project, which helped me apply theoretical knowledge to real-world banking data and understand how meaningful customer insights can be generated through systematic analysis.

In the CSCPA project, I used Python and libraries such as **Pandas, NumPy, Matplotlib, and Seaborn** to clean, preprocess, analyze, and visualize customer-level banking data. I explored customer churn patterns across **France, Spain, and Germany**, along with different age groups, credit score bands, tenure groups, balance segments, and customer activity levels. The analysis also helped identify high-value customers and compare the characteristics of churned and retained customers.

The project strengthened my understanding of **data cleaning, exploratory data analysis, customer segmentation, churn analysis, visualization, KPI development, and data interpretation**. I also developed an interactive **Streamlit dashboard** that allows users to apply filters, view dynamic KPIs, and explore different customer segments and churn patterns.

Along with technical skills, the internship improved my **problem-solving, analytical thinking, time management, documentation, and professional reporting** abilities. I learned how data science can be applied to real-world business problems and how analytical findings can support better decision-making and customer retention strategies.

Overall, the internship at **Unified Mentor Private Limited** strengthened my foundation in data science and motivated me to explore advanced areas such as **machine learning, predictive churn analysis, and advanced analytics**. The knowledge and practical experience gained during this internship will be valuable for my future academic and professional journey in the field of technology and data science.

# REFERENCES


• **Unified Mentor Private Limited.** European Banking Customer Dataset Provided for Internship Project Work.

• **Pandas Documentation.** Python Data Analysis Library.

• **NumPy Documentation.** Scientific Computing for Python.

• **Matplotlib Documentation.** Python Plotting Library.

• **Seaborn Documentation.** Statistical Data Visualization Library.

• **Streamlit Documentation.** Framework for Building Data Applications.

• **European Central Bank.** European Banking and Financial Data Resources.

• **Internship Training Materials and Mentor Support** provided by **Unified Mentor Private Limited**.